



Assignment Cover Sheet

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Submission Date:
21/06/26

Modern Artificial Intelligence

SKIN CANCER CONVOLUTIONAL NEURAL NETWORK

University of Wollongong, Dubai, - Spring 2026

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Abstract

We propose a multimodal skin lesion classification system trained on the HAM10000 dataset, which combines dermoscopic image features with structured patient metadata (age, sex, and anatomical localization) for dermatological triage. Two pipelines of feature extraction were built and evaluated: Pipeline 1: A custom deep convolutional network trained from scratch, and Pipeline 2: A pre-trained EfficientNetB3 backbone fine-tuned on the target domain. The image embeddings from each pipeline were compressed with a supervised feature-reduction neural network, combined with patient metadata for cleaning, and classified with a gradient-boosted XGBoost model trained in the same way in both pipelines to make a fair comparison. On the held-out test set, Pipeline 2 outperformed Pipeline 1 with an accuracy of 81.27% vs 67.94%, a balanced accuracy of 72.08% vs 50.95%, a macro F1-score of 0.6827 vs 0.4721, and a macro-ROC-AUC of 0.9551 vs 0.8941. The results show that transfer learning with pre-trained weights on ImageNet resulted in significantly more discriminative image representations than learning from scratch on a relatively small and imbalanced medical dataset, especially for minority lesion classes. The system is not intended as a standalone diagnostic system, and there are several limitations, such as the use of a single train/validation/test split, the underlying dataset being skewed with lighter Fitzpatrick skin types and the lack of formal statistical significance testing between pipelines.

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[1] INTRODUCTION

Skin cancer is the uncontrolled growth of mutated cells on the epidermis, the upper layer of the skin, often caused by untreated DNA damage from ultraviolet radiation or tanning beds [18]. It is one of the most prevalent malignancies, recording one in every three global cancer diagnosis, knocking on all 7 continents and majority of the countries, the UAE being one of them. The most common and deadly variant is known as Melanoma, reporting 331,722 new cases and 58,667 deaths annually by The International Agency for Research on Cancer (IARC) as of 2022[2].

Skin cancer encompasses a diverse range of subtypes including melanoma, basal cell carcinoma, squamous cell carcinoma, actinic keratoses, benign keratosis-like lesions, dermatofibroma, and vascular lesions, each differing in origin, clinical appearance, and severity [1]. This visual heterogeneity, combined with significant similarity between lesion types, makes accurate classification inherently challenging [3]. The standard clinical diagnostic pathway begins with a visual screening examination, followed by dermoscopic analysis, and ultimately confirmed by biopsy and histopathological examination [3]. Each step in this pathway adds delay, cost, and reliance on specialist expertise, a resource that is severely constrained in many healthcare systems, including those in the UAE [4].

A further limitation of the current diagnostic method is the inconsistency in diagnosis between clinicians. Even experienced dermatologists show discrepancy when evaluating the same lesion: agreement for actinic keratosis and SCC in situ has been measured at only moderate levels ($\kappa = 0.15\text{--}0.67$), while agreement for invasive SCC was found to be poor ($\kappa = 0.05\text{--}0.15$) [9]. This disagreement is not attributable to clinical incompetence, but to genuine ambiguity in lesion morphology and the absence of universal consensus guidelines for borderline presentations [9].

In response to these limitations, artificial intelligence and deep convolutional neural networks (CNNs) have emerged as a clinically validated approach to automated skin lesion classification. The foundational study by Esteva et al. trained a single CNN end-to-end on 129,450 clinical images spanning 2,032 disease categories, achieving diagnostic performance on par with 21 board-certified dermatologists across two critical classification tasks [3]. Subsequent work has reinforced this finding: a comparison involving 58 dermatologists, including 30 experts, found that the majority were outperformed by a CNN-based diagnostic model.[5]. More recently, deep CNN architectures applied to the HAM10000 dermoscopy benchmark, the dataset used in this study, have achieved classification accuracies of up to 94.59%, outperforming both traditional machine learning methods and established pretrained architectures [7]. In a prospective clinical study made by assessing the diagnostic performance of dermatologists cooperating with CNN, it was found that accuracy of diagnosis increased drastically from 74.1% to 86.4%[10].

This paper proposes a CNN-based skin lesion classification system trained on the HAM10000 dataset as a clinical decision support tool for dermatological triage. The system implements two parallel deep learning pipelines, both built on a late fusion architecture that jointly processes dermoscopic images and structured patient metadata (age, sex, and anatomical localization). Pipeline 1 employs a custom CNN trained from scratch, while Pipeline 2 leverages EfficientNetB3 pre-trained on ImageNet, fine-tuned on the target domain. The two pipelines are systematically compared to quantify the performance benefit of transfer learning in this clinical context. As the HAM10000 dataset comprises exclusively confirmed skin lesion images, the system is scoped to classify among seven lesion categories and does not determine whether an arbitrary image is lesion-free [3].

[2] BACKGROUND AND MOTIVATION

2.1. *The Clinical Challenge of Skin Lesion Diagnosis*

Melanoma, although it is the least common lesion type recorded, is by far the deadliest form. According to GLOBOCAN data, only 7.5% of globally melanoma recorded cases are located within Asia, yet the region accounts for 22.4% of melanoma-related deaths, indicating a mortality rate nearly three times higher than its incidence rate [2]. One of the core reasons being its clinical appearance, which is often misdiagnosis as a benign lesion. This showcases

that many patients receive diagnoses at advanced stages, significantly reducing treatment effectiveness and survival chances.

The challenge of melanoma detection extends beyond clinical practice to dataset development. High-quality training datasets such as HAM10000, the dataset used in this report, require extensive validation through histopathology, expert consensus, or long-term patient follow-up, making the collection of accurately labelled images both costly and time-consuming.

2.2. Limitations of Current Diagnostic Approaches

Traditional diagnostic approaches are increasingly unsustainable due to the global shortage of dermatologists and significant disparities in diagnosis. More than one-third of countries have fewer than one dermatologist per 100,000 people, preventing necessary care being provided to patients in accordance [14]. To bridge the gap between these limitations' traditional computer-aided dermatology diagnosis systems were made available, these systems relied on handcrafted features such as the ABCDE criteria, color histograms, and texture descriptors. While these methods demonstrated some diagnostic value, they struggled to generalize across lesion subtypes and were highly sensitive to real-world variations in image quality, lighting conditions, and skin characteristics [3].

These challenges are particularly relevant in the United Arab Emirates, where no national skin cancer screening programme currently exists despite cancer being one of the most impacted states [4].

2.3. The Case for AI-Assisted Diagnosis

Deep convolutional neural networks (CNNs) make diagnosing skin lesions much more reliable. Instead of relying on manual feature engineering, they learn to spot patterns directly from the raw pixel data. This capability is demonstrated in large-scale comparative studies where CNNs achieved higher diagnostic accuracy than the majority of 58 participating dermatologists, including 30 experts. Furthermore, transfer learning has established a clear advantage in skin lesion classification; on the ISIC 2018 dataset, for instance, a pre-trained InceptionV3 model achieved an accuracy of 85.8%, outperforming a custom CNN trained from scratch at 83.2% [5]. Guided by these empirical advantages, EfficientNetB3 was selected as the backbone architecture for Pipeline 2.

Furthermore, Pacheco and Krohling demonstrated that integrating patient metadata including age, sex, and lesion localization, within a multimodal architecture achieved 94.11% accuracy and an AUC-ROC of 0.9426 on the HAM10000 dataset, outperforming image-only approaches [6]. This provides strong justification for the proposed late-fusion multimodal design.

2.4. Problem Statement

This project addresses the various previously listed diagnostic limitations carried by dermatologists across the global healthcare system.

The proposed system accepts a dermoscopic image and patient metadata as inputs and outputs a probability distribution across seven lesion classes. High-risk cases are flagged for expedited specialist review. The system is designed as a human-in-the-loop clinical decision support tool rather than a replacement for clinical judgement.

A key scope limitation is that HAM10000 contains only confirmed lesion images hence does not identify whether the skin is cancerous or not [8].

Project success is defined by achieving:

- A statistically significant performance improvement of Pipeline 2 over Pipeline 1.

[3] EXPECTED IMPACT AND SUCCESS METRICS

3.1. Clinical and Business Benefits

Clinically, the proposed system could help doctors identify malignant skin lesions earlier, speed up diagnosis, and improve consistency between different clinicians. This could be especially useful in local clinics and telemedicine services where access to dermatology specialists may be limited [3].

The impact of the system could be significant because early-stage melanoma has a five-year survival rate of over 95%, while late-stage melanoma has a survival rate below 13% [13]. Therefore, even small improvements in early detection could greatly improve patient outcomes. Previous studies have also shown that doctors achieve higher diagnostic

accuracy when CNN-based decision support systems are used alongside their own expertise, supporting the use of AI as a tool to assist rather than replace clinicians [10].

In addition to the clinical benefits, the system may also reduce healthcare costs. Early-stage melanoma treatment costs around USD 1,800 per patient, whereas late-stage treatment can exceed USD 60,000 [13]. In fact, 88% of melanoma-related healthcare spending comes from only 18% of patients with advanced-stage disease [13]. For healthcare providers such as Cleveland Clinic Abu Dhabi and Mediclinic UAE, the system could reduce unnecessary biopsies, improve efficiency by allowing doctors to assess more patients, and be deployed across multiple clinical sites at a relatively low cost [15].

3.2. Evaluation Criteria

This project aims to compare the performance of two pipelines for skin lesion classification for transfer learning: a custom CNN feature extractor and an EfficientNetB3 feature extractor with XGBoost classification model. The overall accuracy is less informative for the highly imbalanced HAM10000 dataset, and for this reason success is primarily measured using balanced accuracy and macro F1-score.

Performance is assessed based on:

- Transfer Learning effectiveness: if EfficientNetB3 performs better than the custom CNN on the unseen test set data.
- Minority class performance: precision, recall and F1 score for minority classes, including melanoma (mel) and basal cell carcinoma (bcc) which are clinically significant.
- Generalisation capability: consistency of balanced accuracy, macro F1-score, and ROC-AUC across validation and test datasets.
- Feature learning quality: the ability to reduce feature dimensionality while maintaining classification performance.
- Metadata contribution: the impact of patient information (age, sex, and lesion location) when combined with image features.

These criteria collectively contribute to the model performance, robustness, and real-world applicability in medical image classification.

[4] LITERATURE REVIEW

4.1. Relevant Research

Training of Neural Networks was hampered due to small datasets and lack of diversity in dermoscopy images available. To tackle this issue, the dataset used in this project was created by *Philipp Tschandl, Cliff Rosendahl & Harald Kittler* in 2018 for academic purposes under the name “*Human Against Machine with 10,000 images*” shortened to *HAM10000*.

A study conducted by *Esteva, Kuprel, A. Novoa, J.Ko, M.Swetter, M.Blau & A.Thrun* on “*Dermatologist-level classification of skin cancer with deep neural networks*” used 129,450 clinical images consisting of 2,032 different diseases to train a convolutional neural network (CNN). The resulting CNN was tested against 21 board-certified dermatologists on biopsy-proven clinical images. This study establishes that CNNs can be used to train machines on medical images for diagnosing skin cancer lesions.

The pre-existing CNN used in this project was created by *Minxing Tan & Quoc Le*. EfficientNet was primarily designed to optimize limited processing power and data. Unlike traditional CNN architectures that scale only network depth, width, or image resolution independently, EfficientNet introduces a compound scaling method that balances all three dimensions simultaneously. This approach enables the network to achieve higher classification accuracy while maintaining computational efficiency. The EfficientNet family of models demonstrated state-of-the-art performance across several image classification benchmarks while using significantly fewer parameters than previous CNN architectures. Due to its strong performance and efficient use of computational resources, EfficientNet was selected as the foundation model for this project's skin lesion classification system.

4.2. Similar AI Projects

The multi-modal approach applied to this project was already done in 2019 by *Andre G.C. Pacheco & Renato A. Krohling* and published under the paper name “*The impact of patient clinical information on automated skin cancer detection*”. While other models focused on only skin cancer images, this model included tabular patient information such as “localization”, “sex”, and “age” — replicating initial standard procedures dermatologists use — using patient information and skin lesion to diagnose said patient correctly.

To combine two different types of datasets — tabular data containing patient information and images of corresponding patients’ skin lesion — Pacheco and Krohling used existing CNNs (GoogleNet, VGGNet-13/19-bn, ResNet-50/101 & MobileNet) keeping the feature extraction layers and changing the classifier to include patient clinical patient information. Since the number of features extracted from the CNN is far larger than the features coming from the metadata, the resulting features from CNN were fed into a Neural Network for feature reduction and then concatenated with the clinical metadata. This method can be read in detail in their paper.

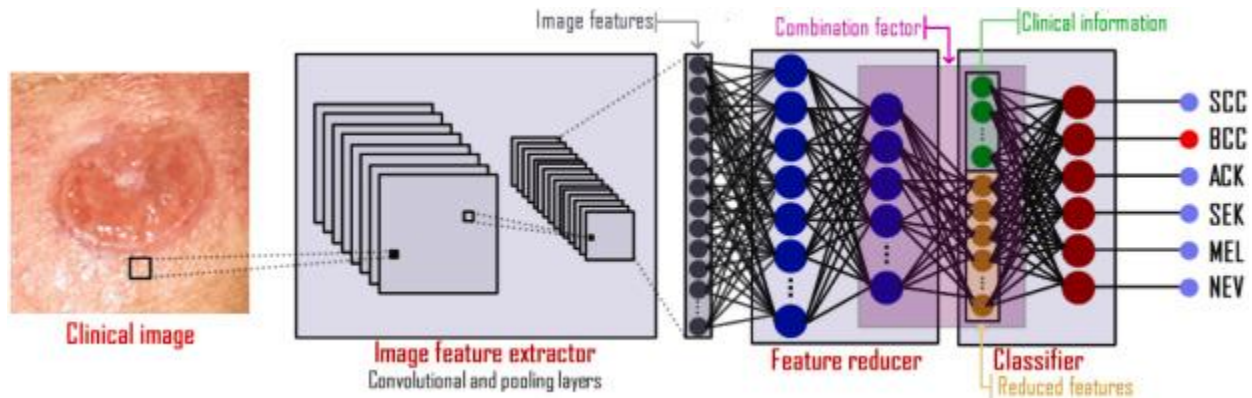


Figure 1. Full pipeline structure for skin cancer detection from Pacheco and Krohling’s work

4.3. Theoretical foundation for selected methodology

The theoretical foundation of this project is based on machine learning and deep learning techniques used for medical classification. The HAM10000 skin cancer dataset contains both dermoscopic images and patient clinical data, thus being suitable for building a multimodal machine learning pipeline for improved diagnostic performance. Convolutional Neural Networks (CNNs) will be used to extract features from the images, then reduced via Neural Networks and concatenated with patient clinical data to form a single master csv file ready to be fed into a classifier pipeline. Primarily, the performance of CNNs will be investigated by utilizing a pre-trained CNN model by comparing it against a model trained from scratch. This project strongly draws upon Pacheco and Krohlings work [6] who demonstrated that integrating image and clinical data can greatly enhance classification accuracy.

[5] TECHNICAL REQUIREMENTS

5.1. System Requirements and Constraints

The computational demands of EfficientNetB3 necessitated high specification hardware to ensure stable and efficient model learning. Although EfficientNetB3 is more compact and smaller than larger variants like B7, it still involves approximately 12 million parameters and processes images at a resolution of 300×300 pixels. It is recommended to run such small to medium models with a VRAM of 16GB to 32GB [17]. To accommodate these requirements, feature extraction was performed on a machine equipped with an Intel i9 Processor, 32GB of system RAM, and a NVIDIA RTX5060 dedicated GPU. Since EfficientNetB3 was going to be used solely for feature extraction, VRAM of 8GB was sufficient. Despite this, the GPU’s VRAM was the real constraint, running EfficientNetB3 with on-the-fly data augmentation risks out-of-memory errors. The RTX 5060 provided extra room to prevent the machine from crashing mid extraction and training. Pipeline 1 (custom CNN from scratch) was trained on a Google Colab T4 GPU instance

(16GB VRAM). The relatively smaller model size (1,281,287 parameters) allowed training with a batch size of 32, completing 17 epochs with early stopping in approximately 45–60 minutes.

5.2. Integration and Scalability Considerations

The current codebase is designed as a stand-alone pipeline and has not been integrated with live clinical systems. For deployment within healthcare, the pipeline would need to be integrated with existing patient management infrastructure and would need to meet the UAE’s medical regulatory requirements for medical AI systems. Access and inclusion of a larger, demographically diverse dataset would be the primary enhancement required before scaling the model in size and integrating medical infrastructure.

[6] DATA & DATASET

6.1. Data Requirements and Availability

The HAM10000 dataset was sourced from Kaggle. The dataset comprises of 10,015 real dermoscopic images across seven diagnostic categories: Melanocytic nevi (nv), Melanoma (mel), Benign keratosis-like lesions (bkl), Basal Cell Carcinoma (bcc), Actinic keratoses (akiec), Vascular lesions (vasc), and Dermatofibroma (df).

Firstly, the classes are heavily imbalanced, nv alone accounts for about 67% of the dataset, while classes such as df and vasc represent less than 27%. This imbalance poses a risk of the model defaulting to the majority-class predictions. Secondly, the dataset has a known demographic skew: images are predominantly sourced from patients with lighter skin tones, reflecting the populations of the contributing clinical institutions in Australia and Austria. This limits the model's generalizability to patients with darker skin tones.

6.2. Data pre-processing

Data pre-processing was done in two structured and reproducible major steps done in two separate python files: `metadata_cleaner.py` and `images_cleaner.py`.

6.2.1. Python file – `metadata_cleaner.py`

As the name suggests, this python file dealt with the cleaning of patient clinical data. The code contains two independent classes called: “*FileImporter*” and “*DataCleaner*”. Using the file path as the only input, the “*FileImporter*” class has the single responsibility of loading the raw metadata csv file into the Python IDE into a variable called “df”.

Following the upload, the “*DataCleaner*” class has 6 private, functional methods which are:

- `_fill_empty_cells` standardize all missing values to NaN
- `_remove_duplicates` drop exact-duplicate rows
- `_encode_sex` binary-encode sex column
- `_impute_age` median imputation for the 57 null ages
- `_label_encode_dx` label-encode diagnosis (dx)
- `_one_hot_encode_localization` one-hot encode body-site location

Before the raw csv file could be subjected to any form of data cleaning, the empty cells present in the raw file were replaced with “NaN” string values (*pandas can recognize “NaN” string as null value*). Notably, only the “sex” and “age” columns had missing values; each column was dealt with differently. The 57 rows with missing “sex” values were dropped whereas the 10 rows with missing “age” values were imputed with the median age of the dataset – 50.

Next, values like “sex” and “dx” were binary and label encoded respectively to: Female = 1, Male = 0 and akiec = 0, bcc = 1, bkl = 2, df = 3, mel = 4, nv = 5, vasc = 6. “dx” was chosen to be label-encoded as it is the target variable. The diagnosis numbers correspond to the index of the probability of the output, thus assigning no ordinal value to the diseases. Lastly, the column “dx_type” was dropped from the dataset as the model is supposed to replicate the steps included in the initial diagnosis of skin cancer. By acting as a proxy for “dx”, “dx_type” can cause data leakage because the type of test will depend on the type of diagnosis the doctor already has in mind.

```

=====
DATA CLEANING PIPELINE – HAM10000 Metadata
=====
Starting shape: 10,015 rows x 7 columns

[1] Standardised empty/whitespace strings → NaN.
[2] Removed 0 exact duplicate rows. (10,015 remain)
[3] Binary-encoded 'sex' (female=1, male=0). 57 'unknown' values set to NaN.
[3] Dropped 57 rows with missing 'sex' values.
[4] Imputed 10 missing age values using median (50.0 years). Nulls remaining: 0.
[6] Label-encoded 'dx'. Mapping: {'akiec': np.int64(0), 'bcc': np.int64(1), 'bkl': np.int64
(2), 'df': np.int64(3), 'mel': np.int64(4), 'nv': np.int64(5), 'vasc': np.int64(6)}
[7] Dropped non-informative ID columns: ['dx_type']

=====
Cleaning complete.
Final shape: 9,958 rows x 6 columns
=====

```

Figure 2. Output after cleaning patient clinical data

6.2.2. Python file – *images_cleaner.py*

The primary function of this file is to enforce image and metadata integrity, i.e. every row has a corresponding image, and every image has a corresponding row and split the cleaned data into train, test, and validation csv files. Like the metadata cleaner file, this cleaner file also has a dedicated, independent file which imports only the image data into the Python IDE by building a dictionary-based image registry.

The Dataset Validator class has three main methods:

- `load_metadata()`
- `verify_images_exist()`
- `clean_dataset()`

These methods combined play a crucial role in maintaining the integrity and consistency dataset before going through the process of machine learning processing. The `load_metadata()` method loads and creates an initial profile of the dataset based on the metadata file. The `verify_images_exist()` method will check the metadata records against the image registry, deleting any metadata records that do not have an image file, and log on to the orphaned image files that do not have metadata records. Finally, the `clean_dataset()` method takes care to ensure a strict one-to-one correspondence between each `image_id` and its related image file by removing duplicate row(s) from the dataset, checking file paths, and verifying that all the image files exist on disk. Further integrity tests prevent duplicate image identifiers and missing file paths from being left in the final dataset. The output is a single clean DataFrame with one row per image with the attributes `lesion_id`, `image_id`, `filepath`, `dx`, `age`, `sex`, and `localization`. The final “df” returned by the `metadata_cleaner.py` contains a total of 9,958 rows and 6 columns with the following distribution:

The Dataset Splitter class is used to create training, validation, and testing datasets. The dataset is divided by the `lesion_id`, not by the `image_id`, so that all the images that belong to the same lesion belong in the same data split. To ensure that the distributions of the classes are similar at each split, Stratified sampling is used, considering the lesion diagnosis (“dx”).

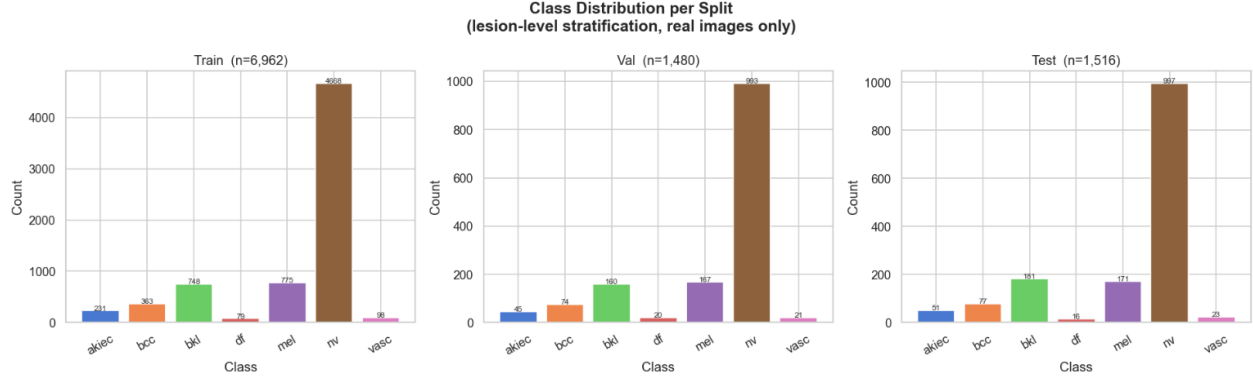


Figure 3. class distribution of train, test, and val data split.

The class splits and then uses a OneHotEncoder fitted only on the training data to encode the “localization” feature. This way we avoided information from the validation and testing sets from influencing the encoding process. Additionally, localization was selected to be one-hot encoded because it is a nominal categorical feature and has no natural ordering by which to label it, as would be required by label encoding for use in neural network training. To guarantee consistent encoding during model inference, the fitted encoder is stored as a class attribute.

6.2.3. Class Imbalance and Data Augmentation

As observed in Figure 4, there is a severe class imbalance; with the class 'nv' containing around 67% of the images, and minority classes like 'df' (115 images), 'vasc' (142 images) being significantly underrepresented. This imbalance would lead to a model that would predict the majority class if left uncorrected. To mitigate this, augmentation was only performed at the feature extraction stage and only for the training split: validation and test images are resized and normalised only, but no augmentation is performed.

Four augmentation techniques were used; these adhere to and are recommended for medical lesion ML models:

- **Random horizontal and vertical flips** ($p=0.5$ for each) were used as the images captured by dermoscopy have no canonical orientation, whereas in natural images a fixed viewpoint can be assumed, and the same image may be captured at a different rotation point across appointments, which increases the viewpoint diversity while having a negligible computational cost.
- Similarly, **full range random rotation** ($\pm 180^\circ$) was performed, because dermoscopic images do not have a “correct” orientation like natural images which are constrained by gravity [19] (Esteva et al., 2017).
- **Minor zoom variation** due to differences in dermoscope placement distance was simulated by randomly resizing the image (scale 0.85-1.0) without cutting away the lesion borders which have diagnostic significance [20] (Perez et al., 2018).
- Lastly, **colour jitter** was limited to brightness and contrast (range 0.8–1.2) to simulate the illumination variability that occurs with clinical devices and operators; hue and saturation were intentionally not jittered because dermoscopic features like blue-white veil or red dots have specific clinical significance that would be lost if hue was distorted [21] (Marchetti et al., 2018).

To avoid 'over-sampling' each minority class to match 'nv', which would involve repeating 'df' about 58 times, and risking the memorisation of the 115 original images, a logarithmic proportional scaling strategy was employed:

$$scale(c) = \log(max_count + 1) / \log(count(c) + 1)$$

The target size for each class was clamped between a lower and upper bound on the augmentation. This method gives higher augmentation multipliers to smaller classes and keeps the relative ordering of the frequencies of classes consistent with the disease prevalence in the real world.

Moreover, a uniqueness constraint was imposed to ensure that each augmented copy of the same image had a different combination of transformation parameters, because if two copies of the same image are augmented in exactly the same way, they will be correlated and will not provide a new gradient signal. This follows Perez (2018) [20], who found that augmentation diversity contributes more to model generalisation than augmentation volume alone.

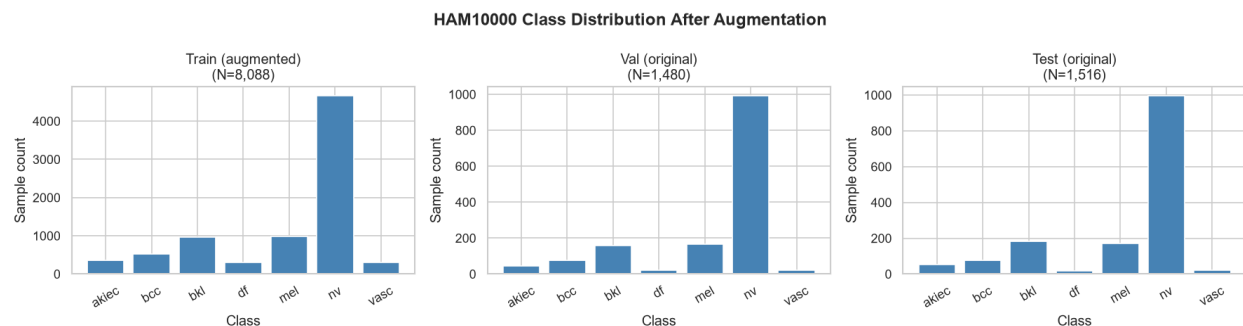


Figure 4. class distribution of train, test, and val data split after augmentation

6.3. Exploratory Data Analysis

Exploratory Data Analysis was conducted as a prerequisite to model design, with the objective of characterizing the structural, demographic, and visual properties of the cleaned HAM10000 dataset prior to feature extraction and classifier training. EDA was performed in two distinct stages: a *Metadata EDA* - examining the tabular clinical, and an *Image EDA* - examining the pixel-level visual properties of the dermoscopic images. It is to note that 57 image files present in the raw image archive were found to have no corresponding metadata record and were explicitly excluded from the image EDA to avoid analyzing unlabeled samples.

6.3.1. Metadata EDA

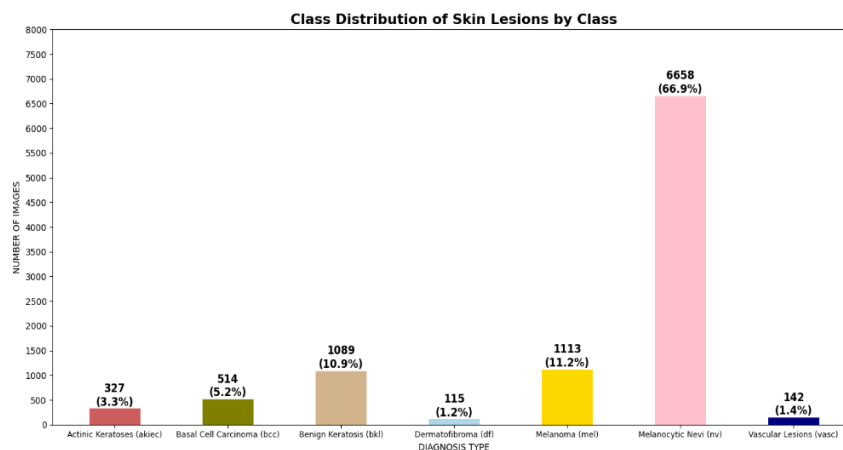


Figure 5. class distribution of entire dataset

Description: A bar chart was constructed using `df['dx'].value_counts()` to display the absolute count and relative percentage of images belonging to each of the seven diagnostic classes. Each bar was annotated with its raw count and percentage of the total dataset to support precise interpretation.

Patterns: The distribution is heavily right-skewed toward a single class. Melanocytic Nevi (nv) accounts for 6,658 of the 9,958 images (66.9%), more than six times the combined total of the three smallest classes. Melanoma (mel) and Benign Keratosis (bkl) form a secondary tier at 11.2% and 10.9% respectively. Basal Cell Carcinoma (bcc, 5.2%) and Actinic Keratoses (akiec, 3.3%) are moderately represented, while Vascular Lesions (vasc, 1.4%) and Dermatofibroma (df, 1.2%) are extremely scarce, together comprising only 257 images.

Analysis & Findings: The chart shows a single class, Melanocytic Nevi, occupying two-thirds of the entire dataset, while the two rarest classes (Dermatofibroma and Vascular Lesions) together make up less than 3%. A classifier trained without correction on this distribution would be incentivised to predict the majority class (nv) for nearly all inputs, since doing so alone yields approximately 67% raw accuracy without learning any genuine discriminative features. This shape indicates the imbalance is dominated by one outlier class, indicating that the dataset is effectively split between one very common category and six comparatively rare ones. The four smallest classes combined (df, vasc, akiec, bcc) still add up to less than the nv bar alone, highlighting how disproportionate the sampling is across diagnoses in the dataset.

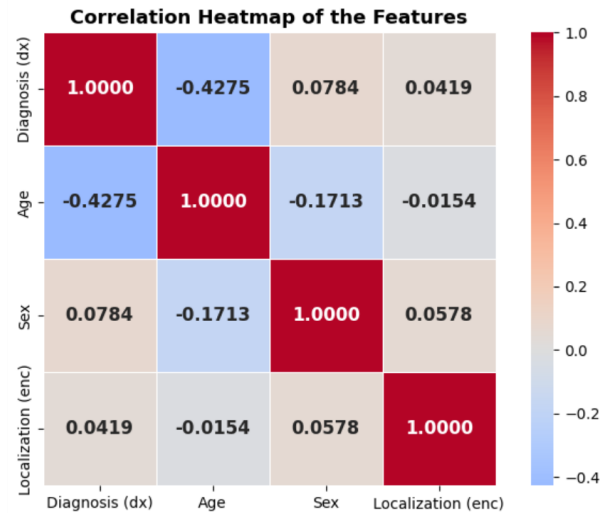


Figure 6. Metadata Feature Correlation Heatmap

Description: A Pearson correlation heatmap was generated across the four numeric features (diagnosis (dx), age, sex, and label-encoded localization) using Seaborn's heatmap function with a diverging colour scale ('coolwarm') centred at zero, allowing simultaneous visual identification of positive and negative correlations and their relative magnitudes.

Patterns: All pairwise correlations across the four features are weak. The strongest relationship observed is between age and diagnosis, at approximately $r = 0.18$. All remaining pairs such as sex & diagnosis, localization & diagnosis, and age & sex; all fall below $|0.10|$ in magnitude, indicating near-negligible linear association.

Analysis & Findings: The heatmap shows one cell that is visually distinct from the rest of the matrix: Age and Diagnosis at $r = -0.4275$, shaded a noticeably deeper blue than any other off-diagonal cell. This is a moderate negative correlation, meaning that as the encoded diagnosis value increases, age tends to decrease, this relationship clearly stronger than any other pair in the matrix. By contrast, Sex & Diagnosis (0.0784), Localization & Diagnosis (0.0419), and Age & Sex (-0.1713) all sit close to zero and are shaded in near-neutral tones, indicating negligible linear association. The colour contrast alone makes clear that age is the one structured feature with any meaningful linear relationship to diagnosis in this dataset, while the rest carry almost none.

6.3.2. HAM10000 Images EDA

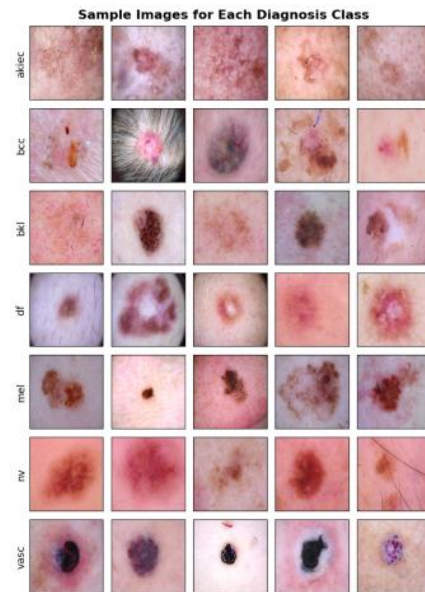


Figure 7. Sample images of each skin cancer class

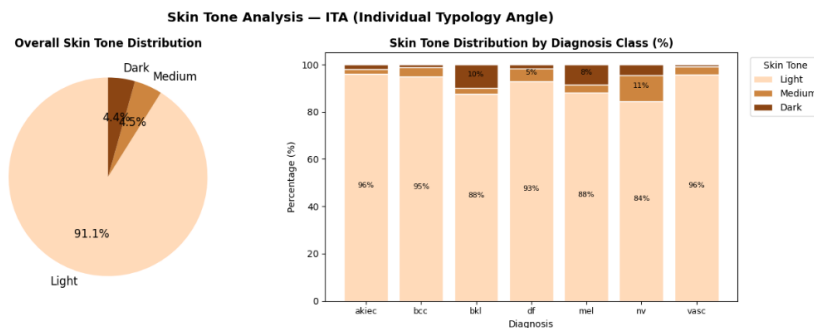


Figure 8. Skin tone distribution

Patterns: The overall skin-tone distribution is heavily skewed toward Light and Medium tone classifications, which together account for the substantial majority of sampled images; the Dark tone category is markedly underrepresented across the dataset as a whole. The per-class stacked bar chart confirms that this skew is broadly consistent across all seven diagnostic classes, with no single class exhibiting disproportionately higher Dark-tone representation relative to the others.

Analysis & Findings: The pie chart shows Light skin tone overwhelmingly dominant at 91.1%, with Medium and Dark almost identical in size at 4.5% and 4.4% respectively, visually the pie has one enormous wedge and two slivers that are barely distinguishable from each other in size. The per-class stacked bars reveal that this skew is not perfectly uniform: bkl stands out with the largest Dark segment of any class at 10%, and mel follows at 8% Dark, both visibly thicker brown segments than the other five classes. akiec, bcc, and vasc sit at the opposite end, each showing Dark segments at only 3 to 4%, barely visible slivers at the top of their bars. nv has the largest medium segment at 11%, a noticeably thicker middle band than any other class's bar. So, while every class is Light-majority, the per-class chart shows real, visible variation in how strongly, bkl and mel are consistently the least Light-dominated of the seven.

A number of other exploratory analyses were done but are summarized here rather than presented as separate figures. As reported above, age was a meaningful difference between diagnosis groups, with the akiec and bcc patients being on average the oldest (66.5 and 66.8 years), and the nv patients the youngest (46.5 years, with outliers below the age of 10). Overall, the ratio of sex was nearly 1:1 (54.3% female, 45.7% male), with a slight male predominance in melanoma and little correlation with the sex of patients with a diagnosis elsewhere, suggesting that sex is a minor auxiliary feature to consider. No combination of these metadata features alone could distinguish lesion types, as evidenced by a pairwise visualization of age, sex, and localization which revealed that there was significant overlap between all seven classes. The mean images per class were computed at the pixel level, and showed the same colour and brightness pattern as the raw images (a dark, concentrated mean for melanoma and a persistent reddish colour for vascular lesions); however, some structural differences were blurred out, resulting in a less distinct separation between nv and mel than in individual images. The dimensions of all of the source images sampled were the same (600×450 pixels), meaning that there were no dimensional outliers and that the acquisition pipeline was consistent. Lastly, the per-image brightness scores confirmed the pixel intensity results, with the lowest median brightness for melanoma and the highest outliers for high brightness for vascular lesions.

[7] METHODOLOGY AND IMPLEMENTATION

7.1. *Selected AI Methodology*

In this paper, two complementary methods of extracting dermoscopic features for diagnosis are compared prior to fusing them with the patient data for final diagnosis. Pipeline 1 is a custom convolutional neural network (CNN) that was designed from scratch. The second pipeline, Pipeline 2, uses transfer learning, using a pre-trained EfficientNetB3 CNN. They were not used as classifiers, but only as feature extractors on the HAM10000 dermoscopic image dataset. Finally, the master csv file (concatenated image features and patient data) was used to train xgboost decision trees to classify the skin cancer lesions. *[image on the next page]*

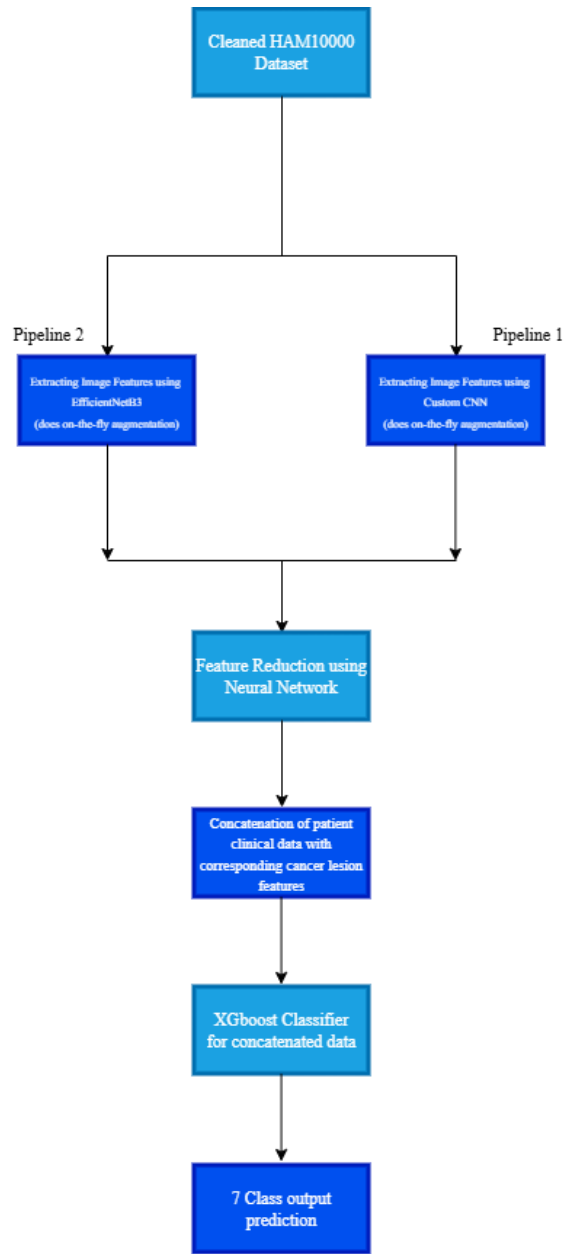


Figure 9. Full Classification Pipeline (from cleaning to extraction to final classification)

The raw embeddings generated by both networks (274 dimensions for the Custom CNN and 1,536 dimensions for EfficientNetB3) were subsequently fed into a separate feature-reduction neural network, which reduced the dimensions to 128 and 64 dimensions for each network, respectively. This reduction step was crucial to ensure a balance between the high-dimensional image features and the relatively low dimensional clinical information. The following table shows how many features are present at each stage:

Table 1: Number of features before and after NN dimensionality reduction for both pipelines

Method	EfficientNetB3	Custom CNN
Number of Initial Features	1536	274
Number of Reduced Features	128	128

7.2. *t*-SNE

t-SNE was applied to the pre-reduction feature vectors extracted by each backbone, EfficientNetB3 and Custom CNN, to visually assess how well each architecture's learned representation separated the seven HAM10000 diagnostic classes before any compression.

The *EfficientNetB3* is significantly better in its class structure. Most of the *nv* class is concentrated in one dense, continuous area with no noticeable overlap with the embedding, and most of the *vasc* class and class *df* form two well-isolated clusters in the upper left corner; both are almost completely isolated from the embedding. The minority highlighted panel also demonstrates that *bkl*, *bcc* and *akiec* are not strictly evenly distributed with *nv* between them but rather tend to cluster together into separate albeit overlapping neighborhoods along the bottom edge of the embedding.



Figure 10. Class distribution on EffNetB3 extracted features

The separation of the *Custom CNN* features is considerably weaker compared to the *EfficientNetB3* features. The *nv* class is divided into several disjointed patches in the embedding rather than occurring in one coherent region, and minority classes, especially minority classes *bkl*, *bcc* and *bkl*, are particularly mixed in with *nv* in the center of the bulk of classes in both panels. Only *vasc* has a clearly separated cluster (bottom left) like it did in the *EfficientNet* embedding, while the other classes share significant overlap with the most common class.

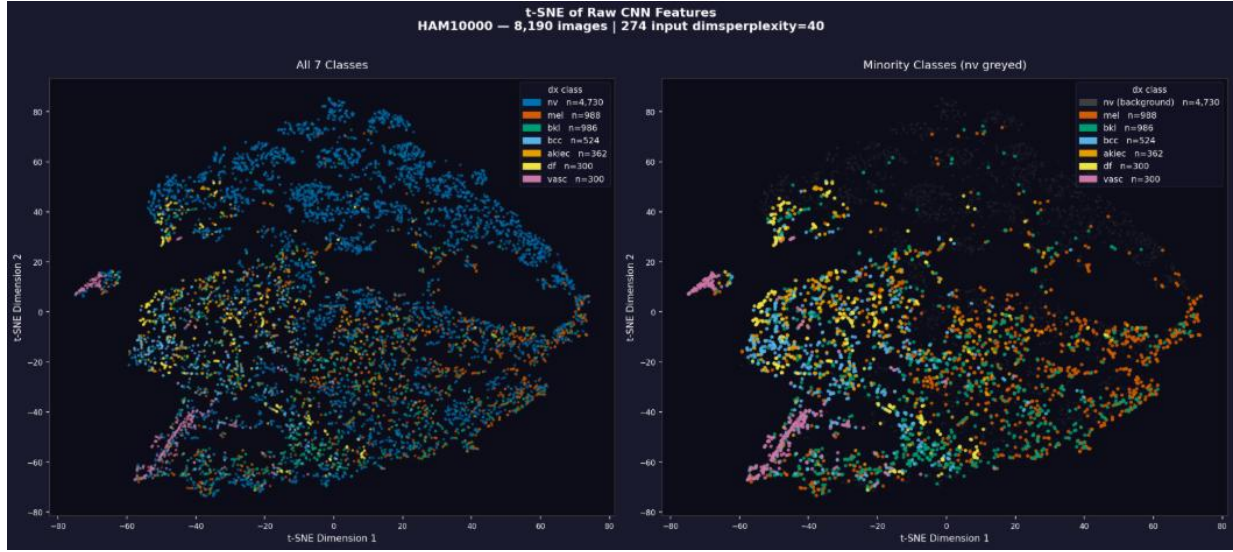


Figure 11. Class distribution on CustomCNN extracted features

This comparison indicates that while the original EfficientNetB3 backbone representation is also learned, it already captures more class-discriminatory structure than the from-scratch CNN's representation. The from-scratch CNN must learn useful representations from the training set of HAM10000, while ImageNet pretraining aids in learning useful representations for EfficientNetB3. The downstream XGBoost performance difference between the two pipelines is informative for understanding the difference in performance between the two pipelines and further emphasizes the significance of the supervised encoder's class discriminative compression step, especially for the CNN pipeline where the initial separation baseline is weaker in terms of visual characteristics.

7.3. EfficientNetB3 Architecture

The EfficientNetB3 model was chosen as the main feature extraction model because of its high classification accuracy and low computational complexity. The network was initialized with pre-trained weights from the ImageNet dataset and then transferred to the HAM10000 skin lesion dataset. This way the model can use previously learned visual features and fine-tune them to the skin lesion classification task.

7.3.1. Model Architecture

In the case of dermoscopic images, the last EfficientNetB3 feature blocks (6–8) were unfrozen and optimized, and the rest of the blocks were left frozen. After fine tuning, the original classification head was discarded, and Global Average Pooling was performed on the last feature maps, which resulted in a 1,536-dimensional feature vector for each image.

These embeddings were not taken from the output layer but from the penultimate layer and are high level lesion characteristics learned by the network. This way, the classification downstream is performed in the learned image representation space, instead of output probabilities, which gives a more discriminative and richer feature space. The feature vectors extracted were then fed into the feature reduction and classification with XGBoost modules of the proposed pipeline.

7.3.2. Training Procedure

Images were resized to 300×300 pixels and normalized with the statistics of ImageNet. For the training images, random horizontal and vertical flips, rotations, zooming, and brightness adjustment were applied, whereas the validation and test images were only resized and normalized.

Adam optimizer with a learning rate of 0.0001 was used to train the model. To avoid overfitting, early stopping was used, and the best model was selected based on the highest validation accuracy. Early stopping was used to prevent overfitting, and the model with the highest validation accuracy was used for feature extraction. The training and

validation curves demonstrated that the pre-trained network was able to fine-tune and adapt to the classification task for skin lesion with training and validation accuracy of around 91% and 85%, respectively.

This results in a 1,536-dimensional image embedding that was further passed to the feature reduction and XGBoost classification phases of the proposed multimodal pipeline.

7.4. Custom CNN Architecture

In contrast, pipeline 1 implements a custom CNN architecture built entirely from scratch using PyTorch, without any pre-trained weights. The implementation is contained in the Jupyter notebook `proj1_fixed.ipynb`, structured to align with the course content taught within CSCI323.

7.4.1. Metadata Feature Transformation

Before the image and metadata branches could be fused, patient metadata had to be converted into a clean numeric feature vector. This was done with a custom *FeatureTransformer* class that implemented the *BaseEstimator* and *TransformerMixin* interfaces from scikit-learn. Internally, it transforms the continuous feature "age" into a standardized continuous feature using *StandardScaler*, and the categorical feature "sex" into three binary features (female, male, unknown) using *OneHotEncoder*. Anatomical localization is already one-hot encoded during preprocessing and represented as binary `loc_` columns in the source CSVs, and is not transformed in any way.

The resulting metadata feature vector is 18-dimensional: 1 dimension for age (standardized), 3 dimensions for sex (one-hot encoded), and 14 dimensions for localization (one-hot encoded). The *FeatureTransformer* is fitted on the training data and then applied identically to the validation and test sets, preventing data leakage.

7.4.2. Model Architecture

The Custom CNN fuses image and metadata inputs via a dual-branch, late-fusion architecture. The image branch takes 224x224 RGB images and passes them through four consecutive convolutional blocks, each consisting of a 3x3 Conv2D, batch normalization, ReLU activation, and 2x2 max pooling, with channels progressing from 3 to 32 to 64 to 128 to 256. The spatial output is then reduced to a 256-dimensional image embedding by an *AdaptiveAvgPool2d* layer.

The metadata branch is composed of two fully connected linear layers (18 to 64 to 32) with ReLU activations and dropout ($p = 0.3$), yielding a 32-dimensional metadata embedding. Internally, the two branch outputs are merged to form a 288-dimensional fusion vector ($256 + 32$), which is passed through a classification head with dropout ($p = 0.5$) and a final linear layer mapping to 7 output classes. In total, the model has 1,281,287 trainable parameters.

For downstream XGBoost classification, features are extracted at the embedding stage: the 256-dimensional image embedding is concatenated directly with the raw 18-dimensional metadata vector, producing a 274-dimensional feature vector that is saved per split as a `.npy` file. This bypasses the metadata branch transformation (18 to 64 to 32) used inside the CNN, instead preserving the raw metadata alongside the image embedding for XGBoost to consume directly.

7.4.3. Training Procedure

The next procedure was used to train the dual-branch architecture, after defining it. Images were scaled to 224x224 pixels, and the training images were augmented online with random horizontal and vertical flips, random rotation ($\pm 30^\circ$), and random color jitter (brightness, contrast, saturation) but not for validation and test images; they received only resize and normalization.

The model was trained with Adam optimizer and a class weighted cross entropy loss function. The class weights were calculated using the scikit-learn's `compute_class_weight` function with the "balanced" strategy, which gives more weight to the classes with fewer samples to reduce the impact of the `nv` class. A `ReduceLROnPlateau` learning rate scheduler was used that decreased the learning rate if the validation loss was not decreasing, and early stopping was used with patience of 8 epochs, stopping the training if the validation loss was not decreasing.

The training was performed on a Google Colab T4 GPU and 2 DataLoader worker processes with a batch size of 32. The best model checkpoint saved to Google Drive was epoch 9 with a validation accuracy of 71.79% and a validation loss of 0.7736; training was stopped early at epoch 17.

7.5. *XGBoost Architecture*

The key pipeline in the project is to use a gradient boosted tree classifier, XGBoost, to learn on top of these previously computed image embeddings, combined with structured patient metadata, without having to train a new neural network.

7.5.1. *Why XGBoost*

XGBoost was chosen as the classifier for the extracted EfficientNetB3 embeddings because of its ability to model non-linear interactions between embedding dimensions and its ability to handle class imbalance, when compared to Support Vector Machines and Random Forest. In addition to the well-established and competitive SVM baseline on frozen CNN features, XGBoost's gradient boosting framework enables direct handling of class imbalance (e.g. by using per-class weights) in addition to the augmentation strategy used during the CNN's pre-processing, which is simply a proportional augmentation strategy instead of relying solely on a single kernel transformation to separate all seven lesion classes.

Random Forest was also considered, but is less likely to do well on fine-grained, visually similar classes (e.g., 'akiec' vs. 'bcc') because the trees it builds have no mechanism to correct errors made by other trees as boosting does.

Furthermore, XGBoost's hyperparameters (learning rate, max depth, subsample ratio, column subsampling) offered fine-grained control to tune XGBoost's capacity against the moderate sample sizes of the minority classes, such as 'df' and 'vasc', reducing the risk of overfitting on the minority classes.

7.5.2. *Data Ingestion*

The pipeline fetches three pre-built CSVs created upstream from the individual merging of reduced CNN (Both EfficientNet and Custom CNN) image-feature vectors with the cleaned metadata table (age, sex, one-hot encoded localization). The merge join generates two redundant label columns (dx_x and dx_y, which is an artifact of the fact that both source tables have a dx column), one of which is kept as the target and the other is dropped, and four identifier and housekeeping columns that contain no predictive signal: file path, augmentation parameters, image ID, and lesion ID. There is no fitting at this step, and it is applied identically and independently to the train, validation and test sets.

After discarding the five non-feature columns, the remaining image-feature dimensions and metadata columns (age, one-hot encoded sex, one-hot encoded localization) are directly converted to a single float32 NumPy matrix (X). No further splitting takes place here as the metadata was already transformed and merged upstream.

7.5.3. *Model and Training Procedure*

The classifier used is 'xgboost'.XGBClassifier(customization= multiclass classification (objective=multi:softprob, num_class=7, eval_metric=mlogloss). The hyperparameters that have been trained are as follows:

- max_depth=8,
- learning_rate=0.03,
- subsample=0.8,
- colsample_bytree=0.8,
- min_child_weight=5,
- gamma=1.0, reg_alpha=0.1,
- reg_lambda=1.0, and
- early_stopping_rounds=30,

With the multiclass log-loss on the validation set as the stopping criterion. The usage of the GPU is detected automatically with a small probe fit (device="cuda") or with the CPU if it is not present.

Unlike the CNN, `compute_sample_weight(class_weight="balanced", ...)` is applied to the rows of the training set, so that underrepresented classes (e.g. df and vasc) will be given a larger weight for every split of the trees than the dominant nv class.

7.6. NN Architecture

The feature reduction stage uses a supervised encoder network implemented in TensorFlow/Keras, which reduces the 1536-dimensional feature vectors outputted by the EfficientNetB3 backbone (or the from scratch CNN pipeline) to 128 dimensional. The architecture comprises of two hidden encoding layers (256–768 and 128–384 units respectively, with automatic tuning of these widths), each followed by a batch normalization layer, a ReLU activation function, and a dropout regularization layer before encoding into the final reduced feature space of a 128-unit bottleneck layer. Rather than manually selecting hyperparameters through trial and error, the encoder's configuration was determined automatically using Keras Tuner's Bayesian Optimization algorithm. The SoftMax classification head attached to the bottleneck is only added to the model as a training signal, predicting the 7 HAM10000 diagnostic classes; this head is removed after training, and the only model used for feature extraction is the encoder.

7.6.1. Why NN

Neural network was selected over the unsupervised alternatives (e.g., PCA or a plain autoencoder) because the diagnosis labels were available, and training the encoder against them ensures that the compressed 128 features were organized to be class-discriminative, and not only reconstructive, providing the downstream XGBoost classifier with a more robust starting representation.

The final model architecture (layer widths, dropout rate and learning rate) were not chosen by hand, but rather by a systematic search using the Keras Tuner's Bayesian Optimization search, where the chosen architecture was evaluated based on validation loss, and using early stopping to prevent overfitting the data.

[8] RESULTS & EVALUATION

Both pipelines were evaluated based on the XGboost classifier at the end, allowing a direct comparison between the features extracted from the custom CNN and EfficientNetB3. Since consistency was maintained in the variables (data augmentation, feature reduction, cleaned data, final classifier) between the two feature extractors, the comparison between them is leveled and as fair as possible.

The difference between the two pipelines is clearly demonstrated in the value of transfer learning in this task. The test accuracy of Pipeline 2 (81.27%) outperformed Pipeline 1 (67.94%) by a significant margin on EfficientNetB3, which was pretrained on ImageNet, on all metrics:

- balanced accuracy (72.08% to 50.95%),
- macro F1 (0.6827 to 0.4721),
- macro-ROC-AUC (0.9551 to 0.8941)

Balanced accuracy and macro F1 indicate that EfficientNetB3's pretrained representations transferred much more successfully to the minority classes than features trained on a relatively small, imbalanced dataset with no prior visual knowledge. This is confirmed by the feature importance results: the EfficientNetB3 pipeline's top 30 predictors were entirely image embeddings, whereas the Custom CNN pipeline used a combination of image and metadata features to make its predictions (see Figure 15), suggesting that transfer learning was able to produce visual features that were rich enough for it to dominate the classification decision without relying on the clinical metadata as an additional signal.

Overall, these findings demonstrate that EfficientNetB3 was able to learn dermoscopically relevant patterns from the large and diverse image corpus that the Custom CNN could not learn entirely from the HAM10000 images. The results of each pipeline are discussed separately and in detail below.

8.1. Pipeline 1 – XGBoost classifier result

The 128-dimensional feature vectors reduced extracted from the custom CNN backbone were used to train an XGBoost classifier. Table 2 summarizes the classification performance on validation and test sets.

Table 2: XGBoost Classification Results – Pipeline 1 (Custom CNN Features)

Split	Accuracy	Balanced Accuracy	F1 Macro	ROC-AUC Macro
Validation	69.26%	54.19%	0.4877	0.9121
Test	67.94%	50.95%	0.4721	0.8941

On the test set, the model achieved an overall accuracy of 67.94% and a macro-averaged ROC-AUC of 0.8941, indicating strong class-level discriminability despite moderate per-class precision-recall trade-offs. The gap between accuracy (67.94%) and balanced accuracy (50.95%) reflects the persistent influence of the majority class nv on overall accuracy.

The confusion matrices in Figures 12 and 13 show that nv achieves the highest recall on both validation (0.80) and test (0.78) sets, consistent with its large sample count. Minority classes df (recall = 0.32 on test) and vasc (0.45) remain the most challenging, though vasc improves markedly on validation (0.73). The bkl class shows consistent confusion with both bcc and mel, likely due to overlapping visual morphology.

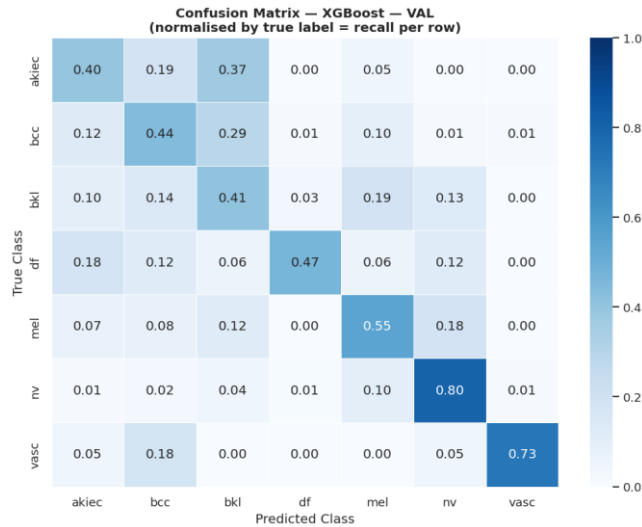


Figure 12. Normalized confusion matrix on the Custom CNN validation set.

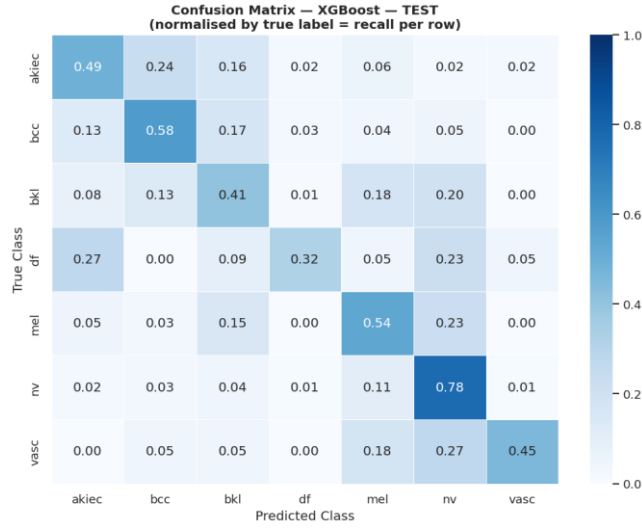


Figure 13. Normalized confusion matrix on the CustomCNN test set.

The ROC curves in Figures 14 and 15 confirm strong separability across all classes, with AUC values ranging from 0.83 (bkl) to 0.93 (bcc, vasc) on the test set. All classes substantially outperform the random baseline (AUC = 0.50), demonstrating that the 274-dimensional CNN embeddings carry meaningful discriminative signal even without fine-tuning on HAM10000.

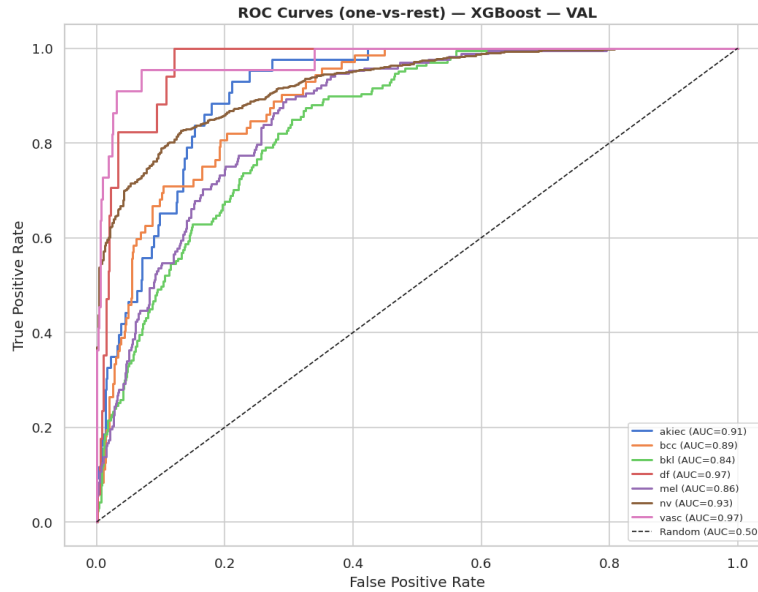


Figure 14. One-vs-rest ROC curves on the CustomCNN validation set.

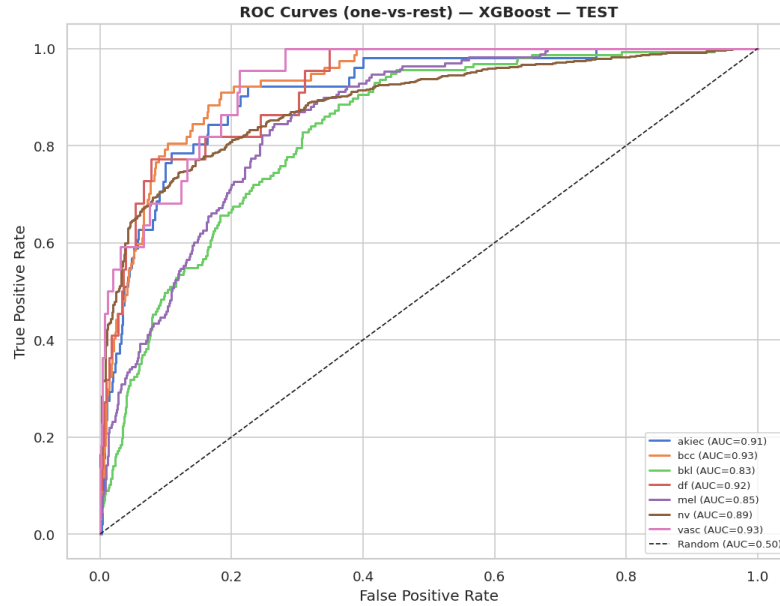


Figure 15. One-vs-rest ROC curves on the CustomCNN test set.

Figure 16 shows the top 30 features by XGBoost gain. Image embedding dimensions dominate (img_106, img_251, img_6), confirming that the CNN backbone captures diagnostically relevant visual features. Notably, several metadata features appear within the top 30 (meta_12, meta_2, meta_0, meta_5, meta_8, meta_10, meta_11, meta_17), demonstrating that patient metadata contributes meaningful predictive signal beyond image features alone and justifies the dual-branch fusion design.

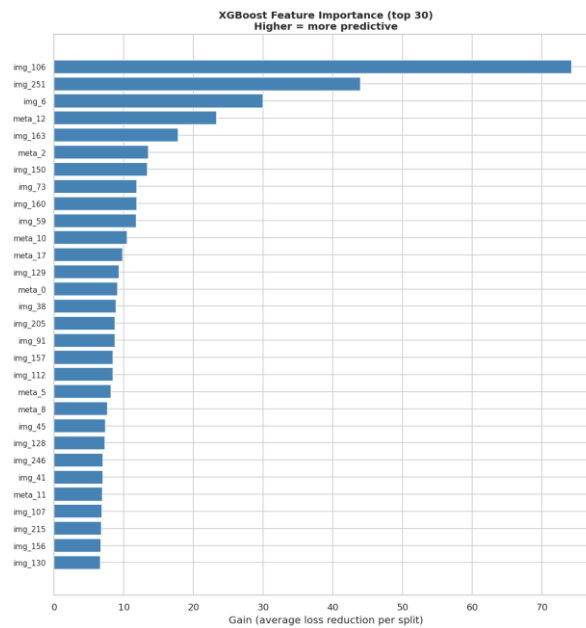


Figure 16. CustomCNN XGBoost feature importance (gain) for top 30 features.

8.2. Pipeline 2 - XGBoost classifier – EfficientNetB3

For pipeline 2, the XGboost classifier converged with early stopping at a best validation log-loss of 0.6162 at boosting round 127. On the validation set, the model achieved:

Table 3: XGBoost Classification Results – Pipeline 2 (EfficientNet Features)

Split	Accuracy	Balanced Accuracy	F1 Macro	ROC-AUC Macro
Validation	81.22%	73.69%	0.6986	0.9442
Test	81.27%	72.08%	0.6827	0.9551

The feature matrices below (Figures 17 & 18) display that both test and validation sets performed similarly. The two classes with notably low recall values are df and mel (mel’s recall increased by 12% in the test holdout). The class df had much fewer data points in its class relative to the other classes, thus explaining the the low recall value. On the other hand, 20% of the time, melanoma data points were classified as melanocytic nevi in the validation set. This can be explained by Damsky and Bosenbergs paper about the complex relationship between mel and nv [22]; this could have caused the machine to get slightly confused between the two skin cancers causing the highest misclassification rate from the EfficientNet extracted features.

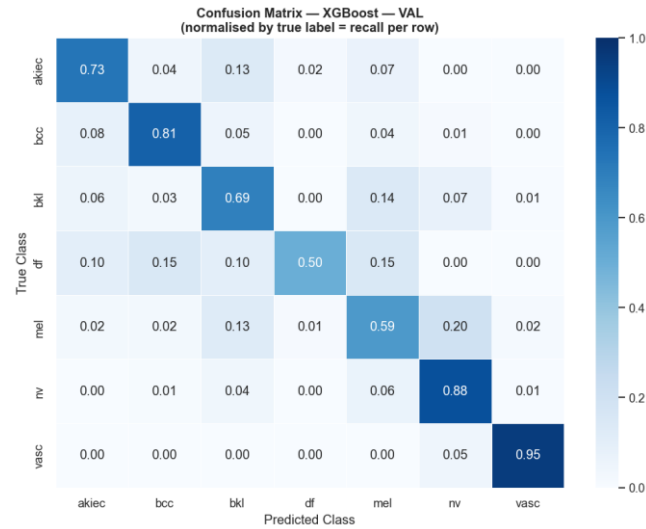


Figure 17. Normalized confusion matrix on the EfficientNet validation set

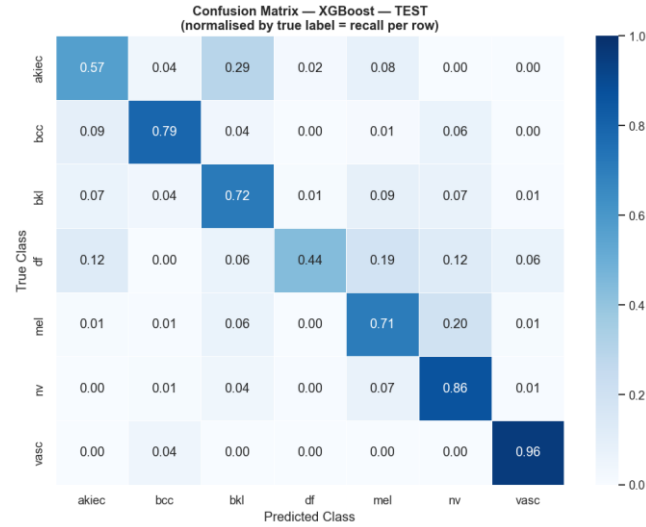


Figure 18. Normalized confusion matrix on the EfficientNet test set

Figures 19 and 20 show one-vs-rest ROC curves across both test and validation splits. All seven classes achieve AUC values of above 0.92, with most classes between 0.95-0.97 indicating that the model separates each lesion type reliably despite the massive class imbalance (data augmentation helped significantly). Although mel has an obvious lower AUC value in the validation set, it improves in the test split, receiving a score of 0.92. The close agreement between the test and validation split ROC curves proves the model has generalized well instead of overfitting against the validation and train split.

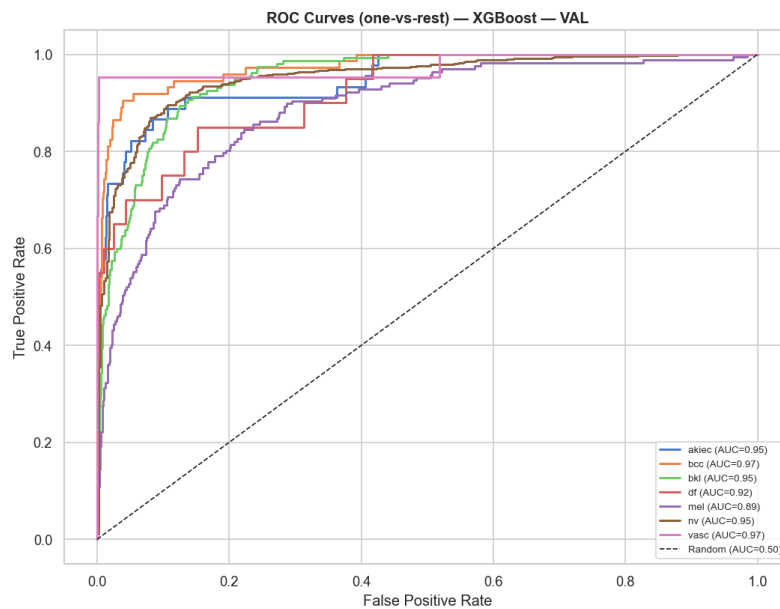


Figure 19. One-vs-rest ROC curves on the EfficientNet validation set.

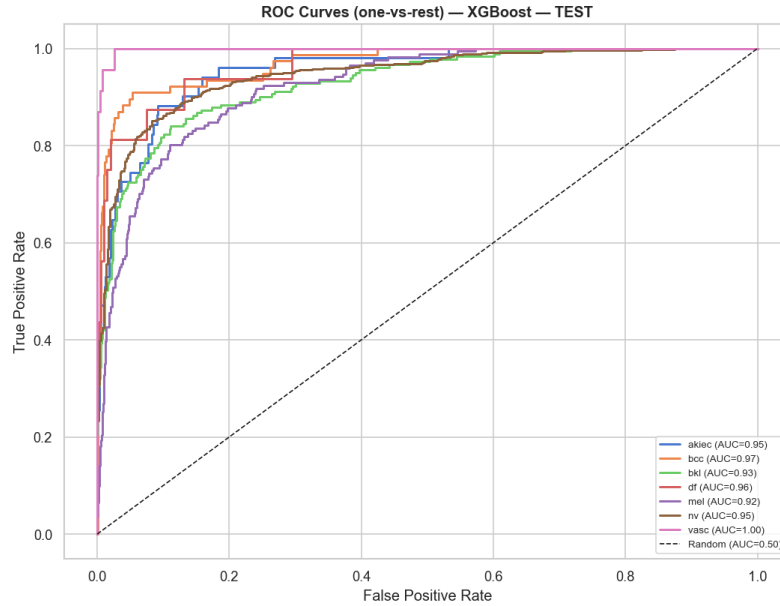


Figure 20. One-vs-rest ROC curves on the EfficientNet test set.

Figure 21 shows the top 30 features that were predictors in the XGboost gain. All 30 predictors featured in the bar graph are image features extracted from EfficientNetB3 indicating that the the ML algorithm extracted powerful predictors from the images. None of the metadata features were primary predictors for skin cancer classification and were likely used as a predictor deeper in the tree, if it was used at all.

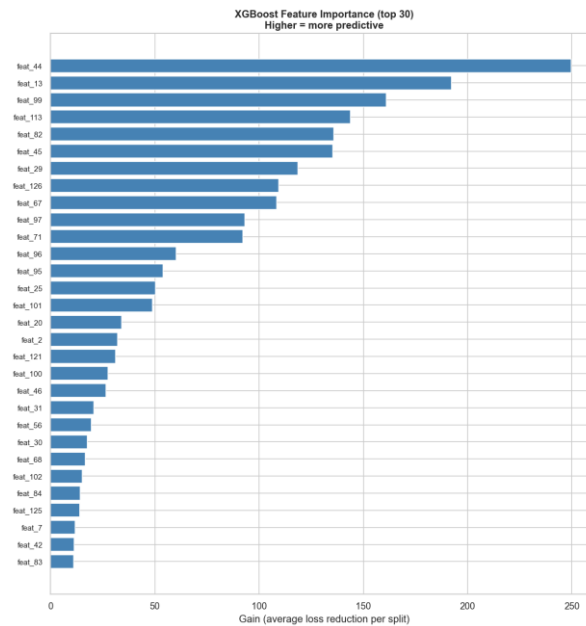


Figure 21. EfficientNet XGBoost feature importance (gain) for top 30 features

8.3. Pipeline Comparison

Table 4 presents a side-by-side comparison of both pipelines across key evaluation metrics.

Table 4: Backbone Architecture Comparison – Pipeline 1 vs. Pipeline 2

Metric	Pipeline 1 (CNN Scratch)	Pipeline 2 (EfficientNetB3)
Parameters	1,281,287	~12,000,000
Pre-trained	No (scratch)	Yes (ImageNet + fine-tune)
Input Size	224×224	300×300
Feature Dim	274	1,536
Parameters	1,281,287	~12,000,000
Training GPU	Colab T4	RTX 5060
Best Epoch	9 (of 17)	5 (of 30)
Optimizer	Adam	Adam
Loss Function	Weighted Cross-Entropy	Weighted Cross-Entropy
Early Stopping	Patience = 8	Patience = 8

[9] CONSTRAINTS AND LIMITATIONS

Throughout making this report various types of constraints and limitations have been observed, starting from the dataset itself. The HAM10000, although widely used for research, contains relatively small data when compared to other modern deep learning systems. It also contains a restricted metadata of only age, sex and localization, other clinical variables could also have played a role in improving the multimodal learning.

The images within the dataset are very biased; some lesion categories contain very few samples; hence the model performance estimates higher variance for these classes.

All experiments were performed on a single train/validation/test split, instead of using k-fold cross validation, because of the cost of repeatedly fine-tuning EfficientNetB3. Thus, the reported metrics are based on a single partition of patients and confidence intervals for the reported F1 and accuracy scores were not calculated. The split may give different results, especially for minority classes like df and vasc, with relatively small numbers of examples and large variations in per-class F1-scores with single misclassifications.

Another drawback is that the data set HAM10000 is very biased towards lighter Fitzpatrick skin types as it was compiled mostly from clinical institutions in Austria and Australia. The Fitzpatrick skin type is not a metadata field in HAM10000, and this study cannot directly quantify the model's performance difference on darker skin tones where dermoscopic colour cues like the visibility of the pigment network are different. This is an unmet fairness and generalizability issue for potential future clinical use, especially in a demographically diverse population like the UAE.

Lastly, although the augmentation strategy outlined above was devised to prevent overfitting minority classes, augmented samples are still transformations of a few original images (e.g. 115 original images for df). The unique-combination constraint reduces but does not eliminate the possibility that the model partially learns to recognize

recurrent properties of the augmentation transformations themselves rather than simply lesion morphology; this is a risk discussed in the Results section above.

[10] FUTURE WORK

Future work should focus in three directions:

- First, with k-fold cross validation, confidence intervals could be reported around per-class metrics, which is of importance for minority classes like df and vasc, where single-split results may not be representative of the true model performance.
- Second, the inclusion of Fitzpatrick skin type (either by further annotating HAM10000 or by enriching the training data with additional demographically diverse dermoscopic data sets) would enable the fairness limitation mentioned to be directly measured and addressed, instead of being acknowledged but not quantified.
- Third, in the image-only vs. image-and-metadata ablation, the contribution of individual metadata fields (age, sex, localization) could be isolated, to understand which clinical variables contribute to the observed improvement, and whether other clinical variables beyond those available in HAM10000 would be worth collecting in a future deployment.

[11] CONCLUSION

The aim of this project was to investigate a specific question: Can transfer learning be useful for skin lesion classification on a small, imbalanced medical dataset and the answer was clear. Pipeline 2 performed better than Pipeline 1 on all of the metrics we monitored, particularly on the balanced accuracy and macro F1 scores, which are least likely to be boosted by the majority of the nv class. This consistency, across accuracy, balanced accuracy, F1, and AUC, is a more convincing result than any single metric would have been alone, and it was a fairly controlled comparison: same classifier, same metadata, same train/val/test split, same augmentation philosophy, with only the upstream feature extractor changed.

The t-SNE visualizations helped us to understand why it happened. The embeddings of EfficientNetB3 showed a clear separation by class before any fine-tuning on HAM10000 was done, whereas the embeddings of the Custom CNN were more tangled even after training from scratch. It was helpful to see that the downstream classification numbers followed this closely, as it was not due to noise in the XGBoost stage, but due to the quality of the representation.

We recognize that a CNN trained from scratch on only ~8000 images will not be able to match a pre-trained backbone, even with identical downstream handling. This is a useful generalizable finding for anyone weighing the cost of pre-trained infrastructure against training from scratch on a similarly modest medical dataset.

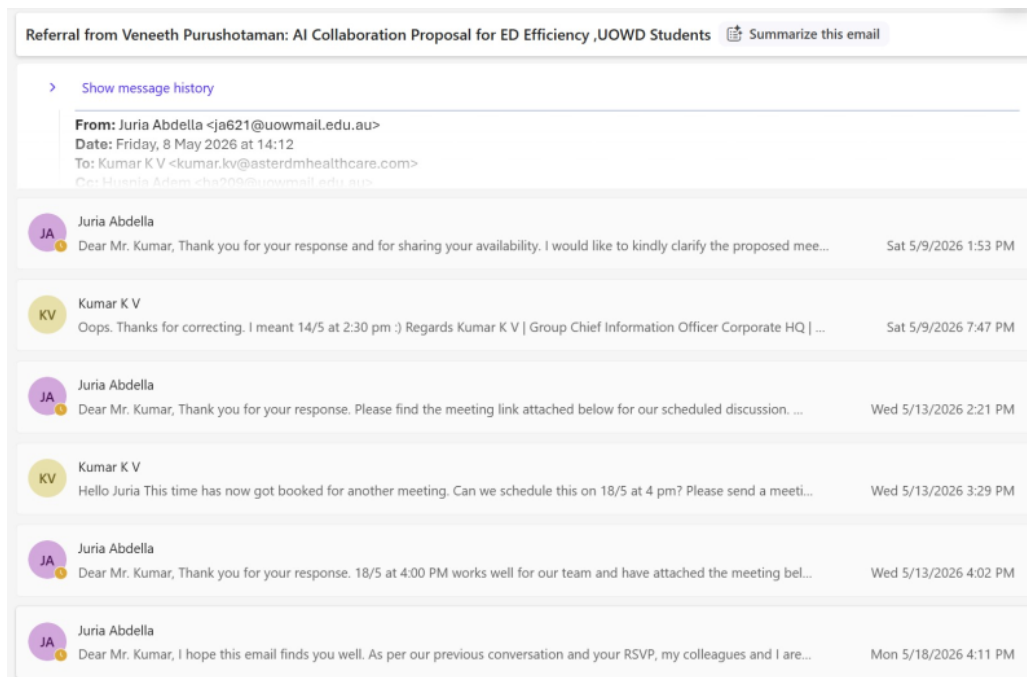
None of this means the work is finished. The comparison was run on a single data split rather than with cross-validation, so we have reported the size of the gap honestly without claiming it has been formally tested for statistical significance, and the dataset's skew toward lighter Fitzpatrick skin types means these results describe performance on the population HAM10000 represents, not a guarantee of how the system would generalize beyond it. We see these less as problems with what we've built and more as the next steps on what can be built: cross-validated confidence intervals, a more demographically representative dataset, and a deeper understanding of which clinical metadata fields contain the most signal would all help to make a result that, even now, makes a fairly strong case for pretraining in this kind of low-data clinical setting.

[12] APPENDECES

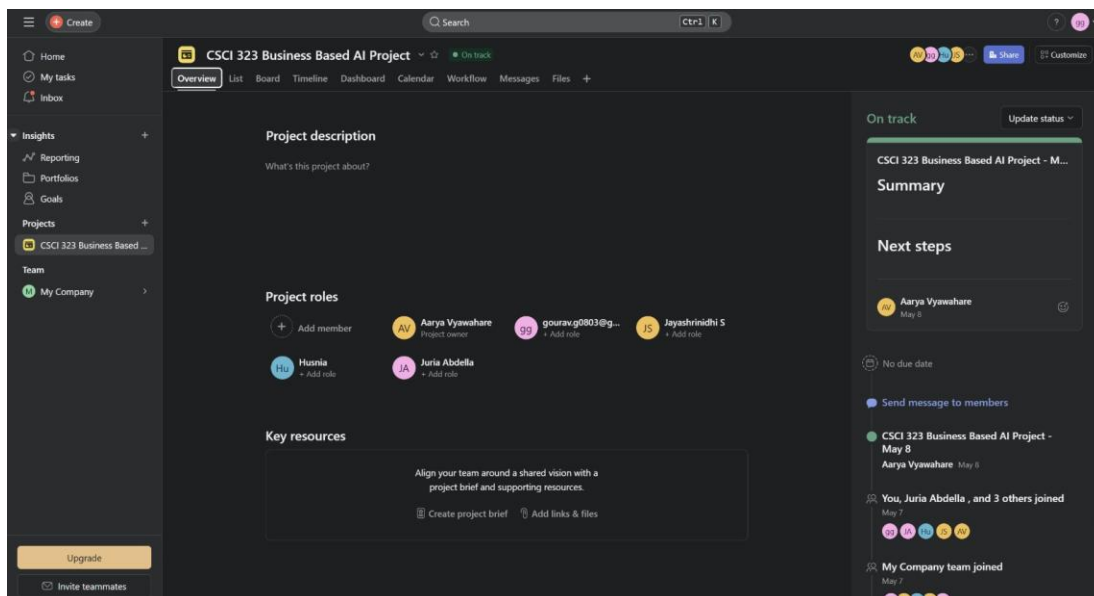
12.1. GitHub Link

https://github.com/areyaya/Skin_Cancer_Classification_HAM10000_CNN

12.2. Miscellaneous Images



Discussion with potential stakeholder.



Project management using Asana

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